

A Probabilistic Early Warning System for Predicting Financial Liquidity Crises in U.S. Banks

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Abstract—The collapse of Silicon Valley Bank and Signature Bank in 2023 resulted in massive economic distress and shockwaves of negative public sentiment. These failures reveal a key limitation of traditional early warning systems: relying solely on financial metrics and binary classification models of economic distress is insufficient for accurately predicting crises in financial institutions. To tackle this limitation, this study proposes a probabilistic early warning system (EWS) to forecast liquidity crises in US intermediate sized banks. In contrast to traditional binary classification models, this proposed model utilizes a continuous risk score, allowing for proactive oversight of banks at the highest risk, as well as a custom curated, multidimensional dataset consisting of accounting, macroeconomic, and non-financial signals. The model uses an Optuna tuned LightGBM algorithm, achieving strong performance metrics with the highest R^2 of 0.9198 and Precision@5 of 0.67, meaning it identifies the riskiest banks with the highest accuracy. A key contributor to this increased accuracy is the addition of macroeconomic and non-financial features which improved the predictive accuracy from 0.131 R^2 with accounting data alone to nearly 0.9 when using all features. These results advance EWS in financial risk forecasting by introducing a more flexible, interpretable, and highly precise model for probabilistic risk estimation.

I. INTRODUCTION

A. Problem Definition

In 2023, the sudden collapse of Silicon Valley Bank (SVB), Signature Bank, and other regionally significant financial institutions sent shockwaves across the economy, culminating in disrupted venture capital markets and emergency responses from regulators. According to traditional financial metrics, these banks displayed healthy tier 1 capital ratios and their loan to deposit ratios (LDR) did not reflect the bank being in a state of high risk [1], making their sudden collapse completely unexpected. Thus, the inability to detect the imminent failure of these banks exposes a critical flaw in existing systems – merely utilizing traditional financial indicators is not sufficient in forecasting and predicting financial liquidity crises.

B. Current Approaches

Looking at the existing literature in the field, Fasano et al. [2] apply deep learning and explainable AI to predict corporate bankruptcy by using over 4 million firm-level financial records. However, one crucial limitation is the lack of non-accounting dimensions such as governance and ESG considerations, limiting the model's robustness.

Additionally, Arias et al. [3] developed a multi-classification model for OECD banks by incorporating regulatory and macroeconomic indicators as well as utilizing SMOTE to generate synthetic data and minimize class imbalance. However, their study utilized a binary variable to indicate a distress event, missing the opportunity to prioritize institutions by degree of stress.

This blind spot among existing predictive models is particularly consequential for intermediate sized banks, managing between 100 - 675 billion dollars in assets, large enough to significantly impact regional economies and financial markets, but small enough to not be classified as Global Systemically Important banks (G-SIBs) or to be subject to the full scope of the post-2008 regulatory requirements since they do not exceed the 700 billion dollar asset threshold set by the Federal Reserve for the most stringent oversight category.

C. Proposed Approach

To address the limitations of traditional early warning systems for financial liquidity crises, this paper proposes a multidimensional early warning system which integrates accounting, macroeconomic, and non-financial metrics. In addition to utilizing conventional financial metrics such as Tier 1 capital ratios, loan-to-deposit ratios (LDR), and liquidity coverage, this model incorporates external, non-accounting indicators such as regional inflation, state-level unemployment rates, large-scale layoffs, and public sentiment signals. Furthermore, rather than utilizing a binary label for distress, this paper opts for a continuous risk score, enabling for a more nuanced assessment of stress levels across banks. By integrating these typically isolated data streams, this proposed system aims to better capture early signals of financial stress and failure, yielding a more robust and context-aware evaluation of liquidity risk.

D. Results Overview

The proposed probabilistic early warning system demonstrates high level performance and was effective across both global evaluation metrics and targeted supervisory use cases. The tuned LightGBM model [5] achieved the highest $R^2 = 0.9198$, the lowest Mean Squared Error (MSE) of 0.0022, and the highest Precision@5 score of 0.67 compared to baseline

models, Ridge Regression [6], XGBoost [7], and Random Forest [8]. These results reveal that the integration of alternative data sources such as non-financial and macroeconomic data as well as shifting from a binary classification of risk to a continuous risk score vastly improve predictive power.

II. BACKGROUND

A. Liquidity Crisis Definition and Modeling

This paper builds upon the theoretical framework of probabilistic early warning systems which estimate the probability of a crisis event occurring. More specifically in the context of this study, the risk of a bank being in a liquidity crisis is modeled as increasing with the number and severity of risk thresholds breached in a given quarter, and banks exceeding 4 or more defined thresholds are assigned higher probabilistic risk scores, reflecting greater vulnerability to a liquidity crisis. The following thresholds were utilized: loan-to-deposit ratio (LDR) greater than 90%, flagged by both the FDIC and IMF as indicative of excessive leverage and insufficient liquidity buffers [9], high-quality liquid assets below 10% of total assets, based on Basel III LCR standards. [10], brokered deposits exceeding 25% of total deposits, considered a destabilizing factor by the FDIC [11], tier1 capital ratio below 8%, matching minimum requirements under Basel III [12], net interest margin under 0.5%, reflecting profit pressure far below healthy norms (typically 2–4%), layoff rate exceeding 5 per 1,000 employees, used in FDIC watchlists as an indicator of internal stress, public sentiment score over 50 on failure-related chatter suggests greater concern regarding the bank’s stability, regional inflation above 5%, flagged by central banks as economically disruptive. [13], unemployment above 7%, used in stress-testing scenarios as a severe economic downturn benchmark [14].

B. Target Variable - Continuous Probabilistic Risk Score

Formally, the target variable is a continuous probabilistic risk score $y \in [0, 1]$, representing the likelihood that a bank enters a liquidity crisis based on the number of financial thresholds breached. To model this risk, in order to preserve both regulatory meaning and statistical learning capacity, we apply a logistic transformation of the number of thresholds exceeded:

$$\text{Risk Score} = \frac{1}{1 + e^{-k(x-4)}}$$

Where: x = number of breached crisis thresholds, 4 = domain-defined action threshold, $k=1.5$ = slope hyperparameter controlling steepness. This approach was utilized because it represents how risk is considered negligible when few thresholds are exceeded; however, risk spikes dramatically at 4 breaches, modeling increased vulnerability. Additionally, applying logistic transformation models allows for greater model flexibility while maintaining domain constraints, allowing gradient-based models to learn latent risk buildup while preserving interpretability.

III. METHODS

A. Data Collection

The dataset was custom curated, and all metrics were recorded quarterly between Q1 2015 and Q1 2025 for every bank in the Federal Reserve Board’s (FRB) large commercial bank list within the price range of 100-675 billion dollars [15]. Financial and accounting metrics were extracted from each company’s respective FFIEC quarterly call reports [16]. All relevant financial statement data (e.g., *loans*, *tier1*, *assets*, *int_income*, *int_expense*, *brokered_deposits*, *deposits*, *liquid_assets*) were parsed from these reports.

Regional inflation and state-level unemployment rates were retrieved via bulk downloads from the Federal Reserve Economic Data (FRED) [17], and organized quarterly. Layoff data was collected through Worker Adjustment and Retraining Notification (WARN) files, accessed using WarnTracker [18]; metrics included the total number of layoffs, the number of workers affected and the normalized layoffs per 1,000 employees.

Public sentiment was captured using Google Trends [19] using scripted queries that combine keywords such as “*bank_name liquidity crisis*” and “*bank_name failure*” for each institution. The data set has 1,230 observations across 34 features structured as a panel data set. The dataset is tabular; however, it also exhibits features of being time based through the Quarter column. All data sources are from publicly available or publicly reportable information. No personally identifiable information was used or inferred. WARN Data were accessed through a licensed aggregation platform (WarnTracker), and sentiment scores were solely based on anonymous search volume trends. All data usage compiled with terms of service and the ethical standard for data scraping, labor reporting, and financial transparency.

B. Preprocessing

All metadata columns that do not contribute to predictive performance, such as *bank_name*, *state*, *region*, and *quarter_label*, were removed to prevent the model from generalizing based on temporal or institution-specific identifiers, thereby reducing the risk of overfitting.

Imputation-related flags (*num_thresholds_exceeded*, *was_imputed*) were excluded from the feature matrix to avoid introducing artificial correlations with the target variable and to prevent confounding effects in SHAP-based interpretability analyses. Additionally, the threshold-based variable *num_thresholds_exceeded* was removed to eliminate label leakage, as it directly reflects downstream outcomes encoded in the target.

Temporal variables were encoded as integers ranging from 0 (Q1 2015) to 40 (Q1 2025), enabling a time-aware validation strategy and mitigating the risk of data leakage from future information. To further reduce look-ahead bias, a *TimeSeriesSplit* cross-validation strategy with $n = 5$ folds was applied, ensuring that training data always precedes validation data temporally.

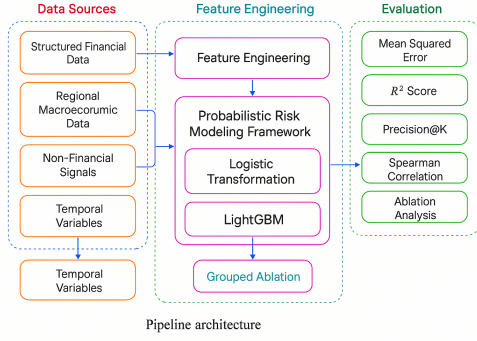


Fig. 1. Pipeline architecture for the proposed early warning system followed by evaluation and benchmarking.

Finally, no explicit feature scaling or normalization was applied, as LightGBM is a tree-based model and inherently invariant to feature magnitudes.

C. Model Architecture

The model is implemented using LightGBM (v3.3.5) [5] which is a histogram based implementation of GBDT, and is particularly well suited for this project due to its proficiency in handling non-linear interactions and discontinuities, preserving interpretability via native SHAP compatibility, and avoiding the need for feature scaling.

Model performance was evaluated on a temporally held out test-set consisting of the final 20 percent of quarters in the dataset (Q1 2023- Q1 2025), ensuring that information from future quarters does not leak into the training data and artificially increase evaluation metrics.

The primary evaluation metric was the mean squared error (MSE), which captures the average squared deviation between the predicted and true probabilistic risk scores. Additionally, the coefficient of determination (R^2) was computed to assess the proportion of variance in the target explained by the model. Higher R^2 values indicate greater success in capturing meaningful variance in crisis vulnerability. This metric is particularly important in probabilistic financial risk modeling, where the target signal may be weak.

During training, the objective function was the MSE due to its compatibility with regression targets bounded in $[0, 1]$. Moreover, MSE serves as a continuous analogue of the Brier score, which is well-established for evaluating probabilistic forecasts, especially given the logistic smoothing applied to the target variable.

IV. RESULTS AND ANALYSIS

To contextualize the performance of the LightGBM model, several benchmark regressors were implemented using the same training and test splits on the identical dataset. The following models were evaluated, each tuned with either default or lightly optimized parameters: Random Forest Regressor, Ridge Regression, and XGBoost Regressor. Among these, the tuned LightGBM model exhibited the strongest performance.

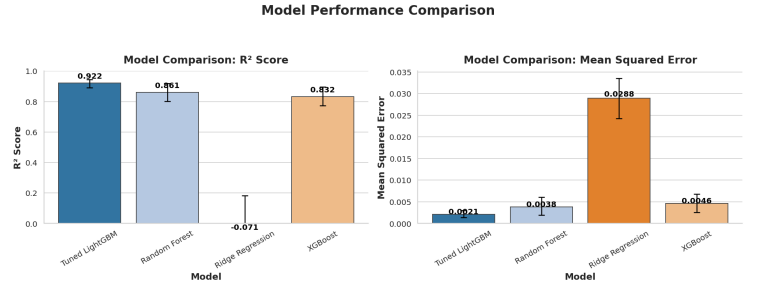


Fig. 2. Model Comparison of global metrics

A. Global Performance

As shown in Figure 2, the regression performance analysis indicates that the LightGBM model achieved the highest $R^2 = 0.922$ and the lowest MSE at 0.0021. In comparison, the Random Forest model yielded $R^2 = 0.861$ and $MSE = 0.0038$, while XGBoost produced $R^2 = 0.832$ and $MSE = 0.0038$. The statistical significance of these differences, with all pairwise comparisons yielding $p < 0.001$, confirms that model selection has a substantial impact on regression quality.

B. Feature Importance

To interpret the marginal importance of each feature in risk prediction, SHAP and feature importance analyses were conducted. To obtain the most reliable feature importance metrics, SHapley Additive Explanations (SHAP) were employed in preference to native feature importance. Native feature importance in tree-based early warning systems, such as LightGBM, tends to overestimate the importance of variables that appear early or frequently in decision splits due to its sensitivity to feature frequency and position within trees. To address this limitation, SHAP values, which are based on cooperative game theory—provide a more robust measure by averaging the absolute contribution of each feature across all predictions.

Figure 3 illustrates that features such as *regional_inflation_rate* and *brokered_to_deposits* rank highly in both LightGBM’s native feature importance and SHAP analysis. However, *liquid_to_assets* appears disproportionately important in the native method despite its low SHAP value. This discrepancy highlights that while certain features may be frequently used in tree splits, their actual contribution to model output is limited, emphasizing the need for SHAP in interpretability workflows.

C. Impact of Feature Categories

To assess the collective value of different feature types (accounting, macroeconomic, and non-financial), a grouped ablation study was conducted. Figure 4 presents separate LightGBM models trained on: (1) accounting factors only, (2) accounting and macroeconomic indicators, and (3) the full feature set including labor market and sentiment-based signals. The transition from accounting-only features ($R^2 = 0.131$) to the inclusion of macroeconomic variables ($R^2 = 0.795$) represents a 664% relative improvement, with bootstrap testing

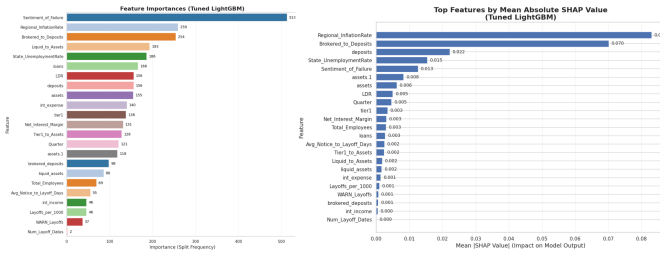


Fig. 3. Feature importance from the tuned LightGBM model based on native LightGBM feature importance versus SHAP summary

confirming statistical significance ($p < 0.001$). Incorporating non-financial variables further improved performance by 9.9%, raising R^2 from 0.795 to 0.893, also statistically significant at $p < 0.001$.

These findings demonstrate that although accounting features are foundational for crisis prediction, they account for only a fraction of the variation in liquidity risk. The inclusion of macroeconomic and non-financial indicators substantially enhances the predictive power of the model.

D. Classification Performance

To evaluate the classification performance of each model in identifying high-risk banking quarters, the Area Under the Receiver Operating Characteristic Curve (AUROC) was computed on the held-out test set. AUROC serves as a threshold-independent metric for ranking quality, measuring a model's ability to distinguish between high and low-risk bank observations.

As shown in Figure 5, the ensemble methods, Tuned LightGBM, Random Forest, and XGBoost, each achieved near-perfect AUROC scores between 0.998 and 0.999, with Random Forest achieving the highest at 0.999. However, despite this marginal performance edge, bootstrap-based pairwise comparisons showed no statistically significant differences between ensemble methods ($p = 0.65$ for Random Forest vs. Tuned LightGBM), suggesting a performance plateau has been reached for this classification task. The tight 95% confidence intervals for ensemble methods (CI width < 0.006) demonstrate strong prediction stability, in contrast to

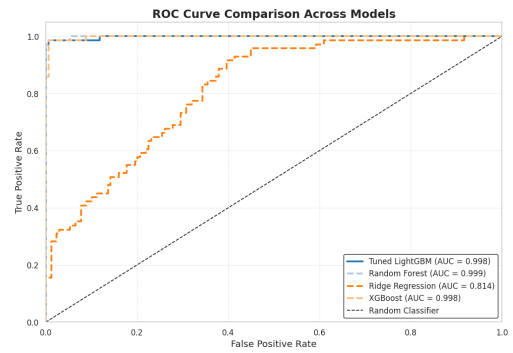


Fig. 5. ROC Curve Comparison to visualize independent classification evaluation across all models

Ridge Regression, which exhibited a wider confidence interval (0.758–0.867), indicating greater variance and the inability of linear models to capture the non-linear structure of financial risk.

To ensure that the near-perfect AUROC of the Tuned LightGBM model is not a result of overfitting, a Precision@5 analysis was also conducted. A perfect AUROC score can sometimes be misleading if the model fails to correctly identify the most at-risk banks. However, Figure 6 depicts the Precision@5 as 0.67, confirming that the model not only ranks observations effectively but also accurately identifies top-risk banks. This supports the conclusion that the model's high AUROC reflects true predictive power rather than overfitting.

E. Global Fidelity Performance and Top-K Precision Analysis

To evaluate how accurately each model preserves the relative ranking of banks based on crisis risk, the Spearman Rank Correlation Coefficient (ρ) was calculated and plotted for all models. Figure 7 shows that the Random Forest model achieved the highest correlation with true risk scores ($\rho = 0.859$). The statistical significance of the difference between Random Forest and Tuned LightGBM ($p < 0.001$) confirms that the ranking quality differences are unlikely due to random variation.

Furthermore, Figure 6 demonstrates that Random Forest outperforms the Tuned LightGBM model in Top-20 precision

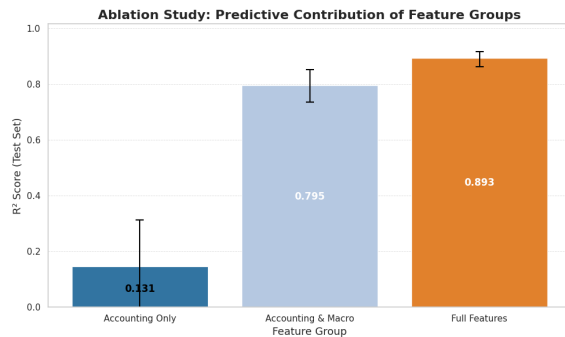


Fig. 4. Grouped ablation study to illustrate the additionally accuracy of adding each feature group

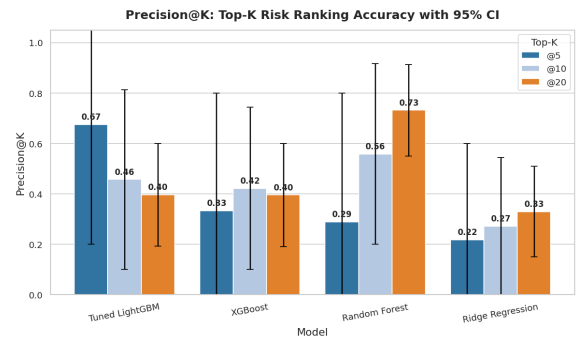


Fig. 6. Precision@K Graph to depict top risk ranking accuracy across all models

Spearman Rank Correlation with 95% CI: Predictions vs. True Risk Scores

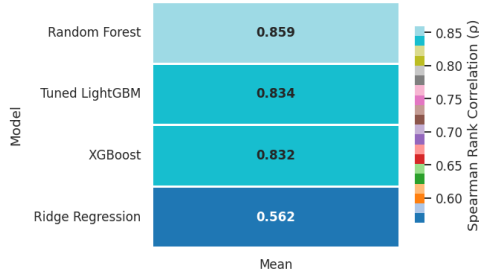


Fig. 7. Spearman Rank Correlation to illustrate global fidelity ranking across models

(0.73 vs. 0.40) with statistical significance ($p < 0.001$). The higher rank correlation and Top-20 precision suggest that Random Forest performs well globally. However, its lower Top-5 precision score (0.29) indicates limited accuracy in identifying the most critical cases.

In contrast, Figure 6 illustrates that the Tuned LightGBM model achieves the highest Top-5 precision (0.67), significantly outperforming all other models ($p < 0.001$). The model's lower Top-20 precision (0.40) but higher Top-5 precision (0.67) further indicates a trade-off, the tuned LightGBM model prioritizes local precision for the highest-risk banks over global ranking performance.

V. CONCLUSION

The proposed EWS outputted superior predictive performance relative to other popular regressors with an R^2 of 0.9198, an MSE of 0.0021, and a Precision@5 of 0.67. These results demonstrate that the model was highly accurate in identifying the highest risk institutions allowing for targeted intervention before a liquidity crisis materializes.

Interpretability analysis using SHAP confirms that macro-financial indicators as well as non-financial signals, with regional inflation, state unemployment rates, and sentiment of failure all being in the top 5 most important features, are critical in risk prediction of liquidity crises, validating the model's alignment to both economic theory and real-world regulatory frameworks.

Ultimately, the findings suggest that utilizing probabilistic machine learning models with multi domain features forms the foundation of a robust and actionable EWS for systemic risk management.

VI. LIMITATIONS AND FUTURE WORK

While the proposed system achieves strong performance, certain constraints and assumptions merit discussion.

Despite the model using a probabilistic risk scoring framework which provides more granularity than binary classifiers, the study assumes static risk thresholds and does not dynamically adapt to economic cycle fluctuations. Therefore, future works could explore adaptive thresholding or regime-switching models that adjust risk sensitivity based on the broader economic context.

Additionally, the study focuses on intermediate-sized US banks. While this segment is under examined and has a large impact if at risk, future research should validate the model's portability to smaller community banks or international banks, especially given differing capital structures and reporting standards. Utilizing a cross-country comparison could determine how well the model generalizes across regulatory frameworks.

VII. ACKNOWLEDGMENTS

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